

Subject Description Form

Subject Code	ME6405
Subject Title	Radiative Transfer: from Theory to Applications
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To understand the importance of radiative transfer processes in energy-related applications. 2. To provide fundamental knowledge of radiative transfer concepts. 3. To enable the formulation and solution of radiative transfer problems using computational tools. 4. To equip students with the computational skills (e.g. Python programming) to design and analyze energy systems involving radiative transfer.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the principles and applications of radiative transfer in energy systems. b. Apply fundamental knowledge of radiative transfer to model and analyze energy systems. c. Design and evaluate energy systems that incorporate radiative transfer principles. d. Develop computational skills to solve complex radiative transfer problems.
Subject Synopsis/ Indicative Syllabus	Introduction: review of basic heat transfer principles: conduction, convection, and radiation; examples of radiative transfer in engineered energy systems and the Earth-atmosphere system. Radiative transfer between surfaces: the electromagnetic spectrum; spectral and directional characteristics of surface radiation; black and gray surfaces; view/shape factors; radiative exchange between black and gray surfaces. Radiative transfer in participating medium: the equation of transfer; gas radiation properties; radiation exchange between gas and enclosure; radiation transmission in glazing and other materials; Monte Carlo method for direct radiative transfer modelling. Atmospheric radiation: atmosphere composition and plane-parallel atmosphere; emission and absorption by gases; the HITRAN database; scattering by gases and particles (aerosols and clouds); atmospheric longwave radiation transfer; atmospheric shortwave (solar) radiation transfer. Case study: radiative transfer in combustion chambers; radiative transfer in water bodies; radiative transfer in solar systems, including photovoltaic systems, concentrated and flat plate solar thermal systems; principles of radiative cooling systems.

Teaching/Learning Methodology	The realization of the intended learning outcomes will be primarily based on lectures under adequate guidance from subject instructors. Tutorials on computational tools and projects will be integrated into the lecture sessions. Students will also be directed to complete a team project with report and presentation to enhance understanding of the subject contents and practice presentation skills. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Tutorial</td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>						Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	✓	✓			Tutorial		✓	✓	✓	Project	✓	✓	✓	✓																		
Teaching/Learning Methodology	Outcomes																																															
	a	b	c	d																																												
Lecture	✓	✓																																														
Tutorial		✓	✓	✓																																												
Project	✓	✓	✓	✓																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignment</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Project</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Examination</td><td>40</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.40 \times \text{Examination} + 0.60 \times \text{Continuous Assessment}$ - The continuous assessment will comprise two components: team project (30%) and individual assignment (30%). The team project and homework are aimed at evaluating their understandings on the physical principles, formulating and solving related problems and enhancing the computational skills. - The examination (40%) will be used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently, and to determine the degree of achieving the subject learning outcomes.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Assignment	30	✓	✓	✓	✓	2. Project	30	✓	✓	✓	✓	3. Examination	40	✓	✓			Total	100 %												
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed																																														
		a	b	c	d																																											
1. Assignment	30	✓	✓	✓	✓																																											
2. Project	30	✓	✓	✓	✓																																											
3. Examination	40	✓	✓																																													
Total	100 %																																															
Student Study Effort Expected	<table><tr><td colspan="2">Class contact:</td><td colspan="4"></td></tr><tr><td colspan="2">▪ Lecture and tutorials</td><td colspan="4">36 Hrs.</td></tr><tr><td colspan="2">▪ Project presentation</td><td colspan="4">3 Hrs.</td></tr><tr><td colspan="2">Other student study effort:</td><td colspan="4"></td></tr><tr><td colspan="2">▪ Self-learning</td><td colspan="4">36 Hrs.</td></tr><tr><td colspan="2">▪ Assignments and projects</td><td colspan="4">50 Hrs.</td></tr><tr><td colspan="2">Total student study effort</td><td colspan="4">125 Hrs.</td></tr></table>						Class contact:						▪ Lecture and tutorials		36 Hrs.				▪ Project presentation		3 Hrs.				Other student study effort:						▪ Self-learning		36 Hrs.				▪ Assignments and projects		50 Hrs.				Total student study effort		125 Hrs.			
Class contact:																																																
▪ Lecture and tutorials		36 Hrs.																																														
▪ Project presentation		3 Hrs.																																														
Other student study effort:																																																
▪ Self-learning		36 Hrs.																																														
▪ Assignments and projects		50 Hrs.																																														
Total student study effort		125 Hrs.																																														

Reading List and References	1. A.F. Mills, C. F. M. Coimbra, Heat Transfer 3rd Edition, Temporal Publishing, 2016. 2. J.A. Duffie, W.A. Beckman, Solar Engineering of Thermal Processes, Photovoltaics and Wind, 5th Edition, Wiley, 2020. 3. G.W. Petty, A First Course in Atmospheric Radiation, 2nd Edition, Sundog Publishing, 2006. 4. O.N. Vassiliev, Monte Carlo Methods for Radiation Transport: Fundamentals and Advanced Topics, Springer Nature, 2017.
Develop Date	December 2024